

RipeSense dashboard demo | updated timing

How to use

Read one sentence for each dashboard section, then spend approximately four to five minutes on Decision Support.

Opening

RipeSense is an offline decision-support application for post-harvest banana ripeness classification from six BME280 sensor readings and literature-derived knowledge-graph features.

Navigation

Use the sidebar in this order: System Architecture, Overview, Data Explorer, Knowledge Graph, Model Results, Interpretability, Robustness, and Decision Support.

Important framing

The application replays saved models on Bath ds_34 data; it is not a live IoT deployment, Brix estimator, shelf-life predictor, or autonomous control system.

System Architecture | 10 seconds

Say

This page shows the five layers: sensor data, literature knowledge, modelling, evaluation, and the Streamlit application.

Overview | 10 seconds

Say

This page summarises the best result, the 13 accepted knowledge-graph rules, the dataset size, and how sensor readings become a ripeness prediction.

Data Explorer | 10 seconds

Say

This page presents the training and test data, five-stage class distribution, sensor relationships, feature ranges, and leakage-safe preprocessing preview.

Knowledge Graph | 10 seconds

Say

This page shows which of the 15 literature rules were statistically validated, which were rejected, and how often each rule is activated.

Model Results | 10 seconds

Say

This page compares sensor-only and KG-augmented Random Forest and XGBoost models, including test metrics, McNemar significance, and confusion matrices.

Interpretability | 10 seconds

Say

This page uses SHAP importance and the rule-alignment score to show which inputs influence the models and how closely learned effects follow the literature.

Robustness | 10 seconds

Say

This page measures performance retention under Gaussian noise, missing values, and simultaneous internal and external temperature-sensor failure.

Decision Support | approximately 4-5 minutes

Do

Open Decision Support, select an input mode and algorithm, then scroll from the prediction comparison through KG features, preprocessing, local importance, fired rules, and storage advice.

0:00-0:40 | Define the two models

Before entering a reading, I will clarify the comparison: the baseline model receives only the six scaled BME280 sensor values, while the KG-augmented model receives those same six values plus transparent features generated from validated literature rules; therefore, the comparison tests whether knowledge-derived context changes the decision, not whether two unrelated models disagree.

0:40-1:15 | Choose the input

I can use manual sliders for a simulated reading, select a labelled row from the held-out test set, or upload a one-row CSV, and I will choose a test-set row so the app demonstrates a real offline benchmark input without pretending that the reading comes from a live IoT stream.

1:15-2:05 | Select the algorithm

I select either XGBoost or Random Forest, and the application then loads the matching baseline and KG pair so both models use the same algorithm family and the only intended difference is the feature representation.

2:05-2:55 | Read the prediction comparison

The agreement banner tells me whether the two models predict the same stage, and the two result panels then show each predicted ripeness stage, confidence, and probability distribution across stages one to five, making it possible to discuss both the decision and its uncertainty.

2:55-3:35 | Inspect KG features

The KG feature table is generated from this exact sensor reading and exposes rule flags, risk scores, and the total violation count, so the knowledge graph contributes inspectable evidence rather than an unexplained hidden adjustment.

Knowledge-Integrated Supervised Learning for Post-Harvest Banana Ripeness Prediction

3:35-4:10 | Check preprocessing

The preprocessing trace shows every raw sensor value, its scaled value, its accepted range, and the same training-fitted scaler used by the experiment, which demonstrates consistent inference and guards against data leakage.

4:10-4:40 | Explain local importance

The local feature-importance tables identify the strongest tree-based features for this particular prediction on the baseline and KG sides, while the global SHAP charts remain available on the Interpretability page for overall model behaviour.

4:40-5:00 | Finish with advice

The fired rules and storage advice translate the KG evidence and predicted stage into practical guidance such as controlling temperature or humidity, but the output remains advisory and does not claim Brix, shelf-life, autonomous control, or rule-based replacement of the classifier.

Closing | 10 seconds

Say

RipeSense combines a strong sensor classifier with transparent knowledge-derived evidence, while the evaluation honestly reports that the KG improves auditability more clearly than statistically significant accuracy.